Jordan curves, the best embedding we can get for K_6 is the one shown in figure 15.15, which still has 3 intersections.

Nevertheless, using arcs allows us to choose a simpler and more regular layout for the vertices.

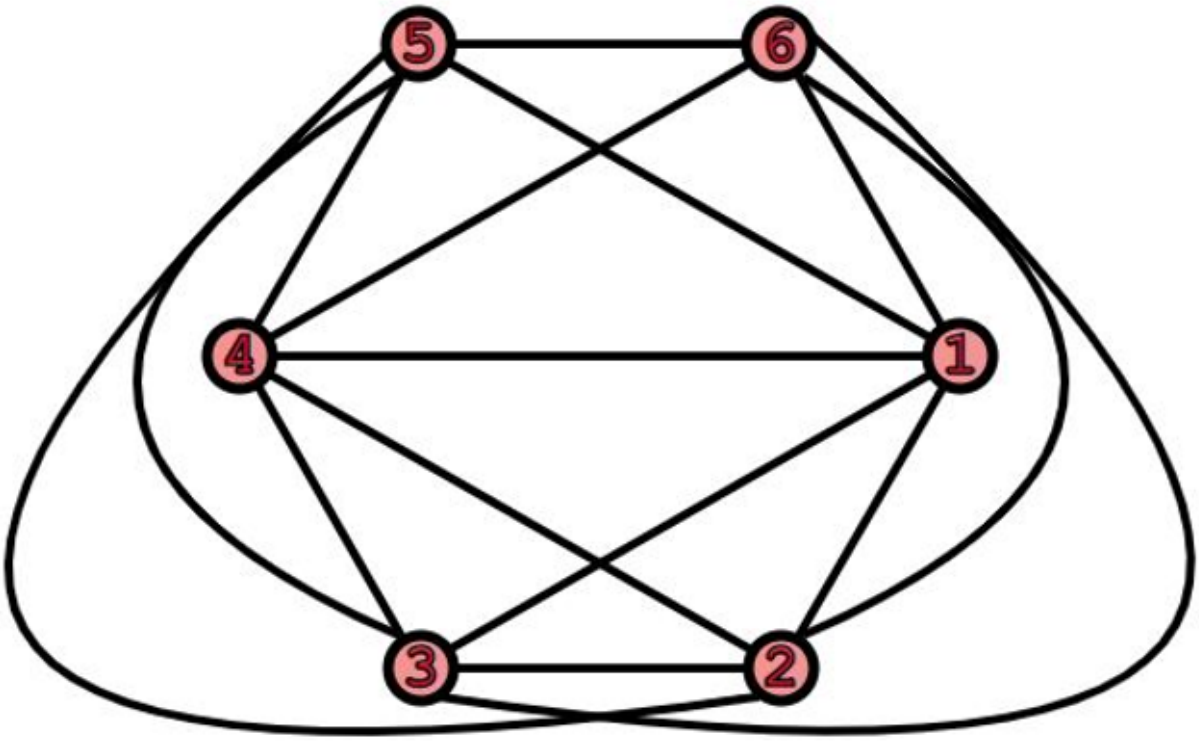


Figure 15.15 An alternative minimum crossing embedding for K_6: using curves for the edges allows us to keep a more regular layout for the vertices.

So, long story short, if we draw large graphs using only straight-line segments, chances are that we have to accept a larger number of edge intersections than actually possible, because rectilinear crossing number is larger than crossing number for many graphs. Does that mean we have to dismiss this way of drawing graphs? Not at all!